"final" Lecture(1) → exam এ এসব থাকবে না শুধু topic থাকবে।

FINAL

"Pin description"

30/03/2026. Sunday

Active high(A)
- 1 (Active)
- 0 (Inactive)

Active low($\overline{A}$)
- 0 (Active)
- 1 (Inactive)

Address ← AD → Data

① $\overline{RD}$ → Read
- 0 (Read)
- 1 (Not read)

② $\overline{WR}$ → write
- 0 = write
- 1 = not write

③ ALE → enable
- address
- Latch
  - 1 (AD0 – AD15 ⇒ address)
  - 0 (AD0 – AD15 ⇒ not address)

④ M/$\overline{IO}$ → input output
- memory
  - 1 (memory operation)
  - 0 (I/O operation)

exam এ এসব থাকতে পারে

M/$\overline{IO}$
- 0 (memory)
- 1 (I/O)

⑤ $DT/\overline{R}$ → Receive

Data transfer

- → 1 (Data transfer)
- → 0 (Data Receive)

⑥ $\overline{DEN}$

Data ↙ ↘ enable

- → 0 (AD0 – AD15 ⇒ Data)
- → 1 (AD0 – AD15 ⇒ not data)

⑦ Ready = 1 (processor currently নতুন কাজের জন্য ready আছে)

= 0 (processor অন্য কাজে busy এখন নতুন কোনো কাজ করতে পারবে না)

CamScanner

Topic: Bus timing Diagram

cycle

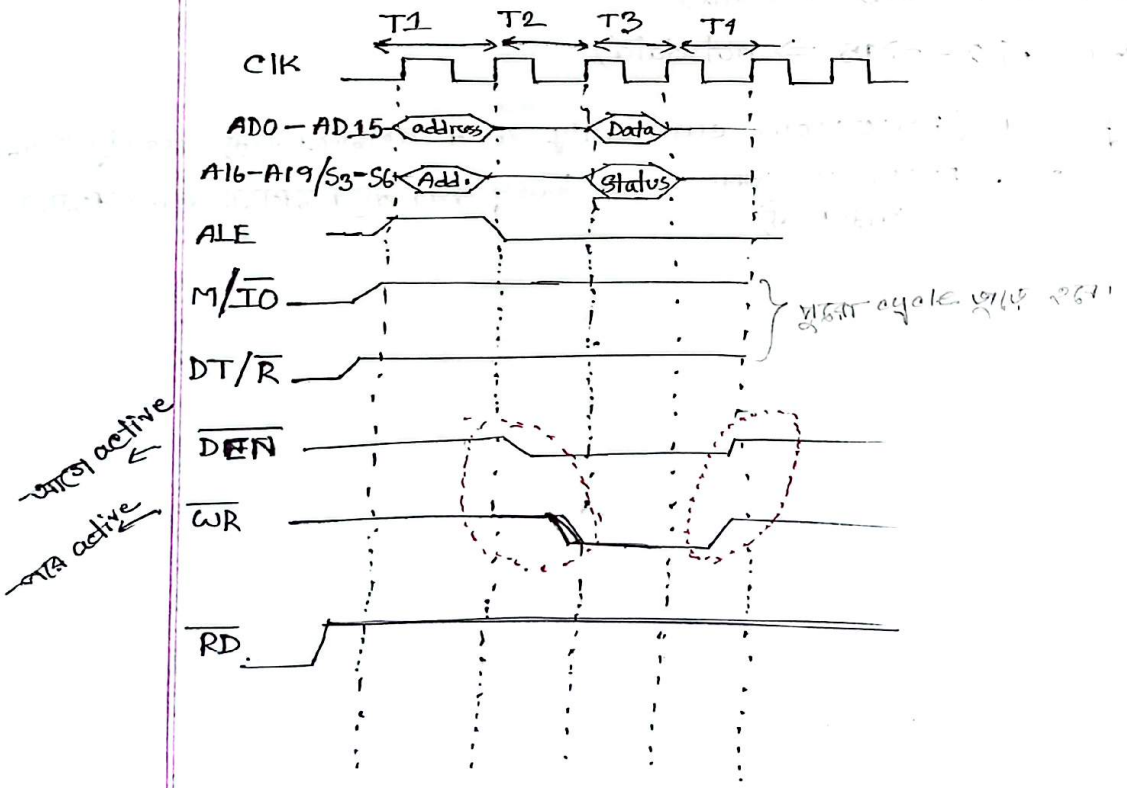
T1 : 1. Send address
2. ALE
3. M/$\overline{IO}$
4. DT/$\overline{R}$

T2 : i) $\overline{DEN}$
ii) $\overline{RD}$ or $\overline{WR}$

T3 :
i) Send Data

T4 :
i) Reverse the operations of T2

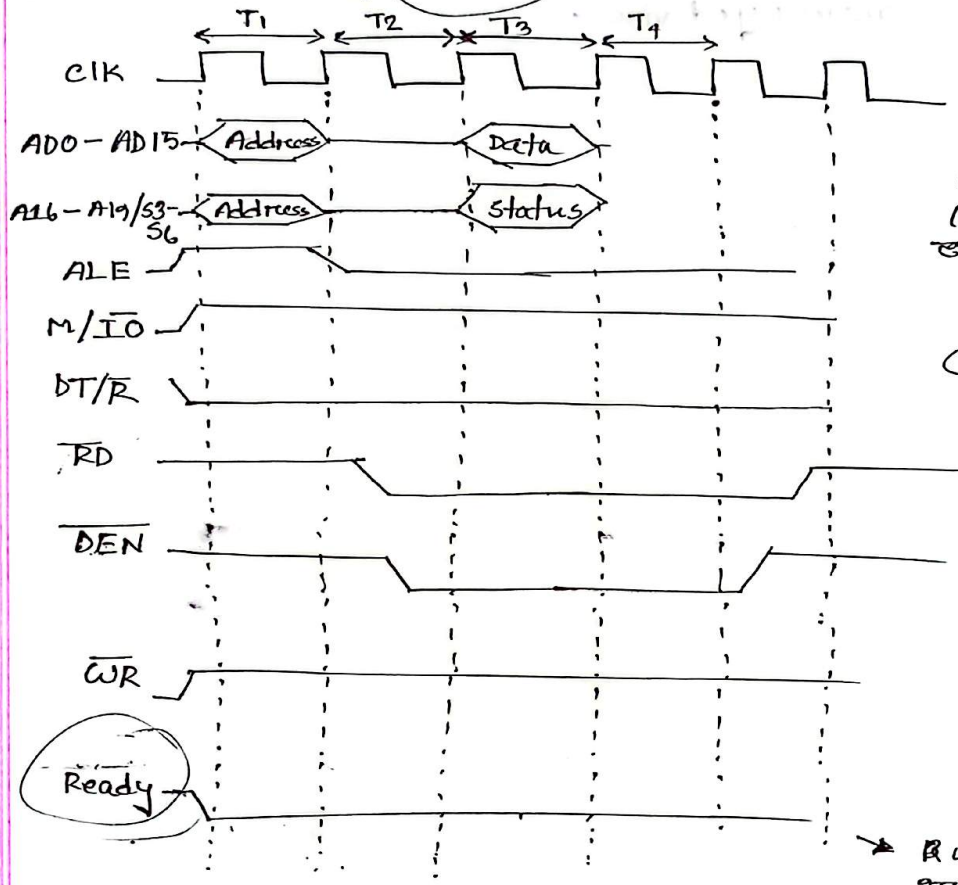
Question: Memory write (→ Data transfer) operation এর জন্য timing diagram draw করো।



lecture -02

Sep 2 (wednesday)

Question: Memory Read

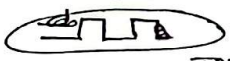


Write → DT

Read → R

Address গুলো বলে দেয়া থাকলে signal আকারে লিখতে হবে।

0101



এটা আকারে Diagram টি।

Read operation এর সময় $\overline{RD}$ আগে Diagram এ বসাতে হবে তারপর $\overline{DEN}$।

আবার write operation এর সময় $\overline{DEN}$ আগে তারপর $\overline{WR}$।

Exam এ pin গুলো change করে দিতে পারে like $\overline{DEN}$ থেকে DEN।

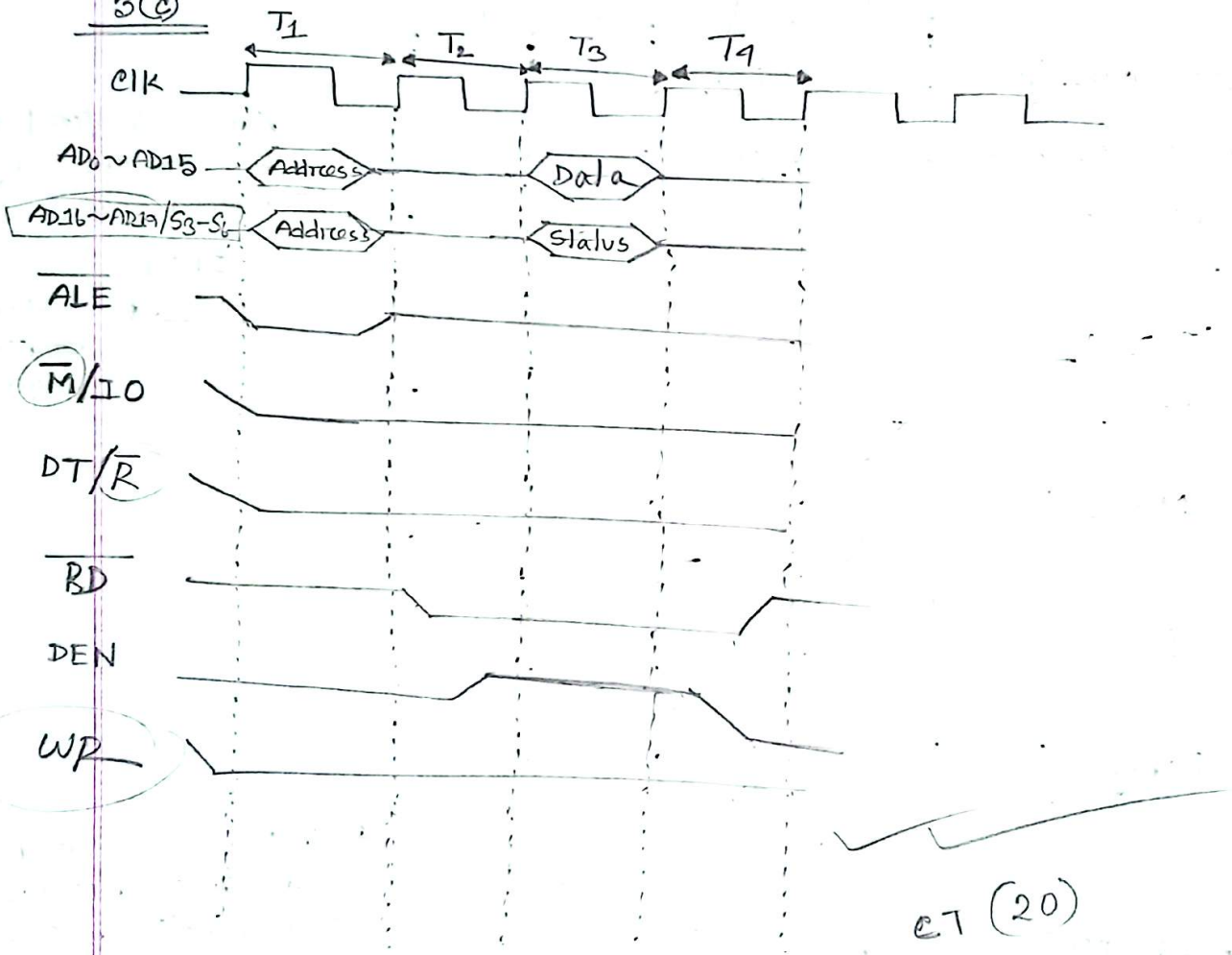
→ এই কাজটা কি নির্দিষ্ট ভাবে হবে? না কারন ready pin inactive. Ready pin active pin. এটাকে off করে রাখলে সেই কাজটা দেয়া হয়েছে সেটা করতে পারবে না। তাই Ready pin কে active করা লাগবে। Diagram এ Ready pin active করা লাগবে।

$\overline{DEN}$, $\overline{WR}$
$\overline{RD}$, DEN

M_write
M_read
I/O_write
I/O_read

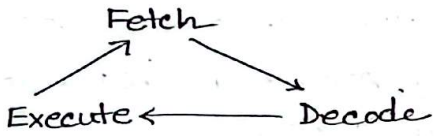
(M→ Read) = কারণ RAM থেকে Data retrieve করছে। RAM মানে memory device.

3(c)



CT (20)

Topic : Pipelining



Bus interface unit → Fetch
Execution unit → Decode, Execute

1) Memory Read operation
2) Jump instruction

এই দুইটার জন্য basic pipelining possible না।

mov AX, 4H ⇒ AX = 4H

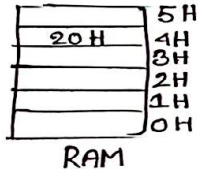
mov AX, [4H]

[4H] → Address

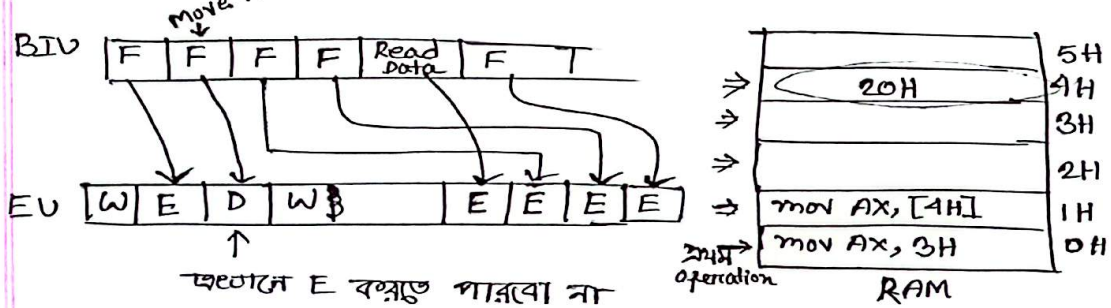
Third bracket দিলে address হয়।

AX = 20H

[4H] address এ যাওয়া। ঐ address এ যেই value আছে সেটা AX এ বসাও



Memory Read operation

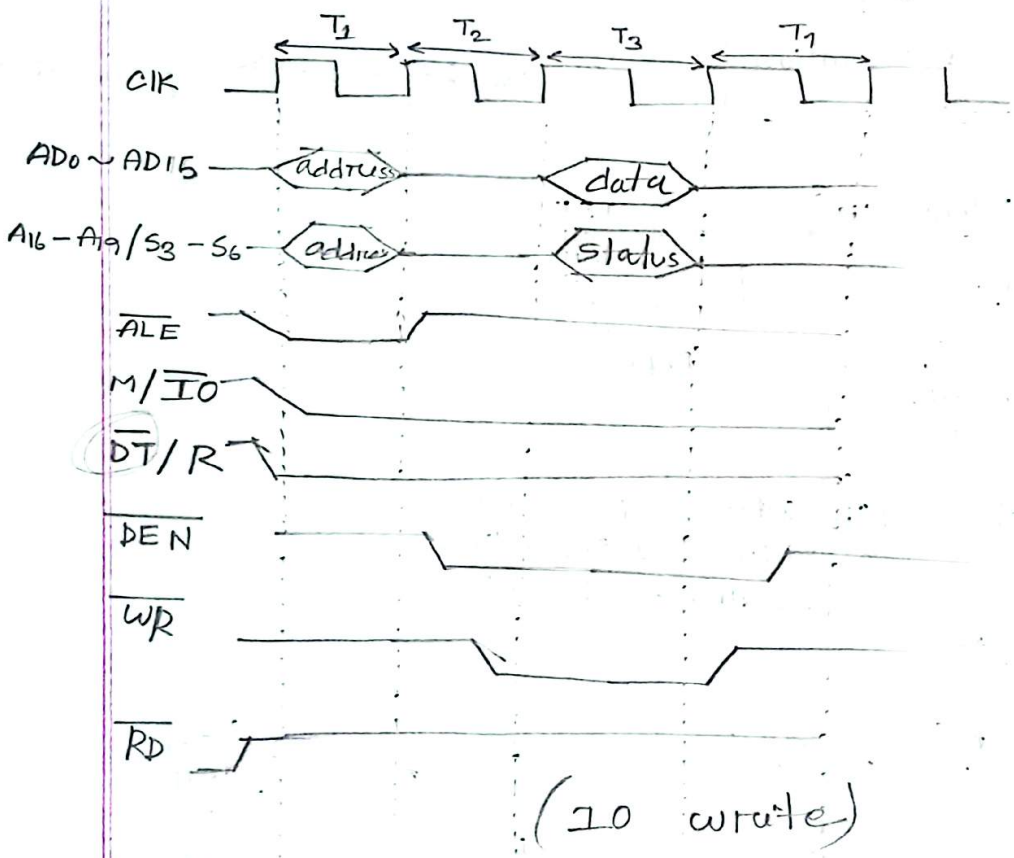


এখানে E করতে পারবো না কারণ [4H] এখনো Fetch করা হয়নি। তাই Decode করা লাগবে। তারপর wait করবো যতক্ষণ না পর্যন্ত [4H] কে fetch করা হয়।

practice !!

CT-03

(1) Draw the timing diagram for IO write operation of microprocessor '808x' showing the activities of address status and Address Data pins (A19-A16/S6-S3), AD15-AD0), $\overline{ALE}$, $\overline{DEN}$, $\overline{RD}$, $\overline{DT}/R$, $M/\overline{IO}$, $\overline{WR}$ in each clock cycle.

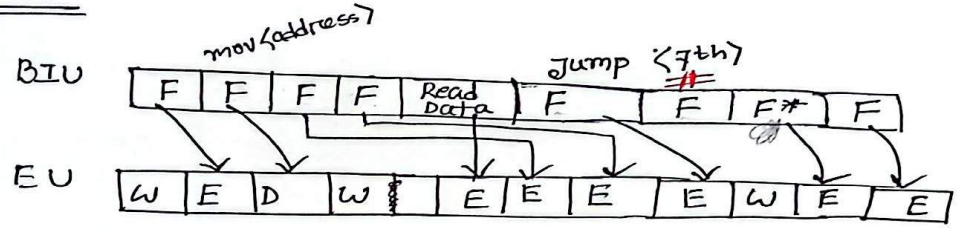


(IO write)

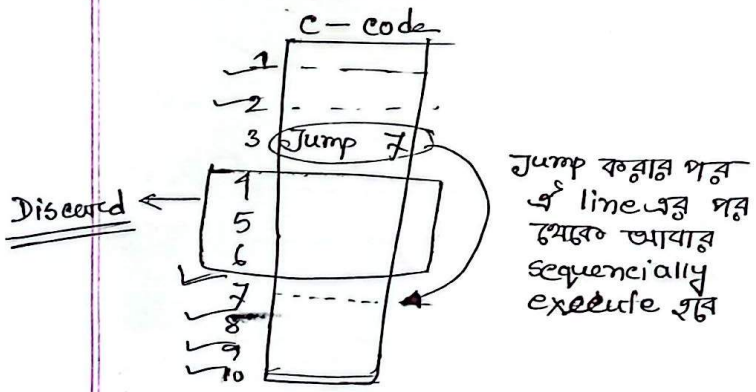
lecture – 03
06 – 09 – 2026
(Sunday).

Move <address> = ~~store~~ memory read.

Slide – 15



Jump instruction



(*) 2nd ins → Jump <5th>

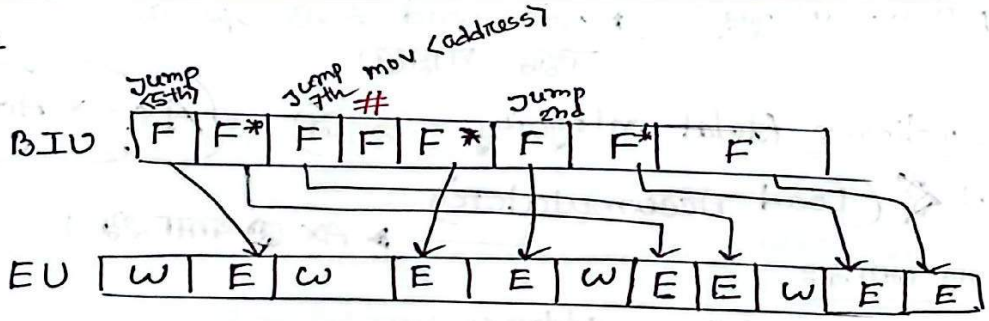
↳ যেই instruction কে jump করতে হয় সেখানে * দেয়া থাকবে.

↳ Discard বুঝাতে "#"

↳ 2nd instruction এ Jump <7th> থাকলে সে Jump করে 7th থেকে execute করত। এর মাঝে যা-ই থাকুক না কেন Discard হয়ে যেত।

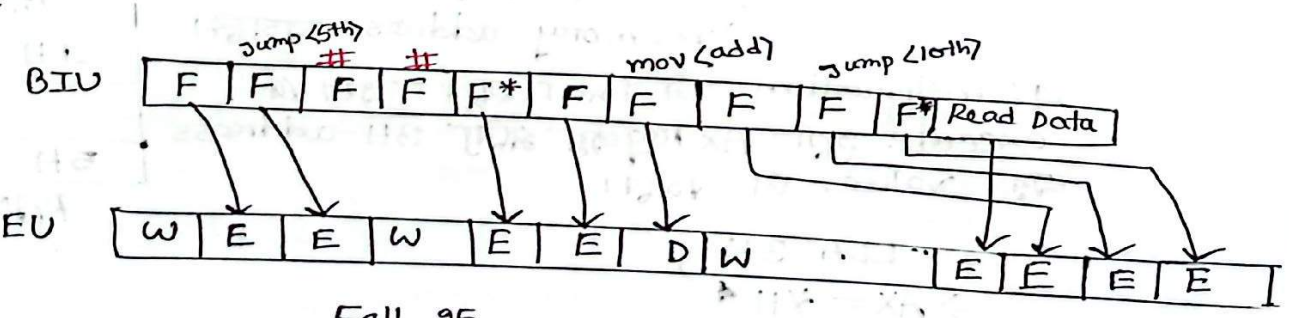
"Summer 2025"

3.@



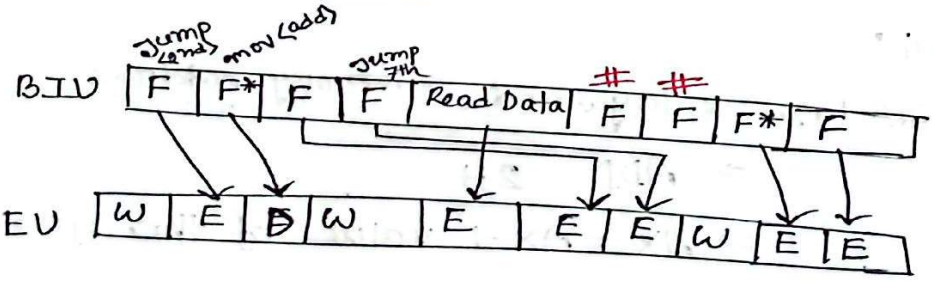
"Spring 26"

3.@



Fall 25

2.@



CamScanner

important for final (10 marks)

Topic : Simple As possible Computers.

lecture - 04 (online class)

Computer-টা খুব → Add and SUB operation করতে পারবে।

Instruction : (total category = 5 টা)

AX → Accumulator

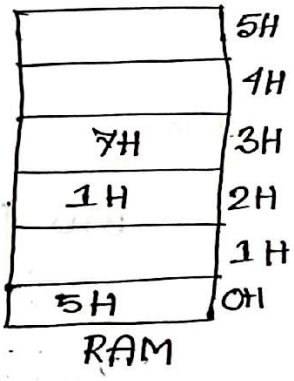
1) LDA (Load Accumulator) → AX কে বলা হয়।

লেখার নিয়ম :

LDA ..... address বসে

LDA 3H

⇒ Keyword (LDA) then memory address বসবে।



এই instruction টা নিয়া গেলে এটা execute হয়ে AX রেজিস্টার মধ্যে 3H address এর value টা বসাবে।

LDA 3H

⇒ AX = 7H

(2) ADD

লেখার নিয়ম : keyword memory address

⇒ ADD 2H

⇒ AX = AX + value of the given address.

* AX এর current value এর সাথে address টার value যোগ হবে।

⇒ AX = 7H + 1H

⇒ AX = 8H

এখন যদি কেউ ask করে AX এর value কত! তাহলে updated

value টা বসাবো। [AX = 8H]

③ SUB

SUB : OH

⇒ keyword (SUB) memory address (OH)

⇒ AX = AX − value of the given address

⇒ AX = 8H − 5H

⇒ AX = 3H

④ OUT ⇒ It displays the value of AX in the Binary Display.

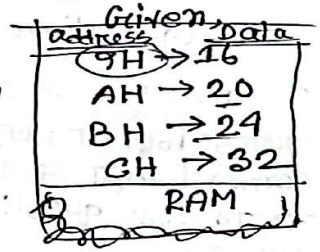
Display : 4H

⑤ HLT (HALT) ⇒ shuts down the computers.

Question : category 1 ——— X ———

① 16 + 20 + 24 − 32 ?

Solution : প্রথমে 16 num. টাকে আনা লাগবে। RAM Table এ check করবো 16 num টা কোথ address এ আছে।



Operation	AX
LDA 9H	16
ADD AH	16 + 20 = 36
ADD BH	36 + 24 = 60
SUB CH	60 − 32 = 28
OUT	28 (out command দিলে AX এ যেই value থাকবে তা display তে চলে যাবে)
HLT	0 (HLT command দিলে computer off হয়ে যাবে)

operation গুলা যেভাবে চালানো হয়েছে সেভাবে code লিখলেও হবে কিন্তু নিচের Question টার জন্য পারবে না।

Code + RAM table দেখা অনুযায়ী চালানো হবে operation এবং execute করার পর result গুলি যেই register এ থাকবে।

(*) RAM table এ Data গুলো Direct decimal এ দেয়া নাও থাকতে পারে। সেইক্ষেত্রে অবশ্যই না মেলে decimal এ নিয়ে নিতে হবে।

Question: Category 2.

Problem 2: Given,

operation: LDA 9H
ADD AH
ADD BH
SUB CH
OUT
HLT

RAM table

Address	Data
9H	32
AH	16
BH	20
CH	24

ADD / SUB এর প্রথম num টা রাখা হয় AX এ 2nd num রাখা হয় BX এ।

computer এর কাজ টা কী ছিল?

Solve:

	AX	BX	Output Reg
LDA 9H	32 ---------->	0	0
ADD AH	32+16 = 48 --->	16	0
ADD BH	48+20 = 68 --->	20	0
SUB CH	68 −24 = 44 --->	24	0
OUT	44 ---------->	24	44
HLT	0 ---------->	0	0

OUT এর সময় AX এ যেমন update হয় না তেমনি BX এও update হয় না। 0

∴ operation : 32+16+20−24 (Ans)

out command output reg এর value update হবে। AX এ যা থাকবে তাই এখানে বসবে।

Question: Category − 3 : Computer operation গুলো ADD, SUB টা দিলেই হুবহু output এভাবে চিনে না। এগুলোকে binary তে/machine code এ convert করা লাগবে। ∴ Machine code এ convert করার জন্য question এ op code এর table দেয়া থাকে।

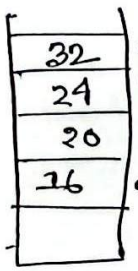
	op code
LDA	0000
ADD	0001
SUB	0 010
OUT	1110
HLT	1111

LDA 9H → 0000 1001

simply binary তে convert করে দিবো

ADD AH → 0001 1010 → A

Lecture - 05
9/09/2026.



16 + 20 + 24 − 32

output Reg		AX	BX
0	LDA 9H	16	0
0	ADD AH	16+20 = 36	20
0	ADD BH	36 + 24 = 60	24
0	Sub CH	60 − 32 = 28	32
28	OUT	28	32 ~~28~~ 0
0	HLT	0	

Machine code ↳ convert

LDA 9H → 0000 1001

ADD AH → 0001 1010

OUT = 1110 XXXX

HLT = 1111 XXXX

op code	
LDA	0000
ADD	0001
SUB	0010
OUT	1110
HLT	1111

↳ চেষ্টা থাকবে।

Question Solve :

Spring 26

(4) initially, AX = 5H
BX = 10H

$(-5+2X9A-3^2)_{16}$

$=(-5+9A+9A-(3X3))_{16}$

$=(-5+9A+9A-(3+3+3))_{16}$

$=(-5+9A+9A-3-3-3)_{16}$

2X3 = 2+2+2
= 3+3
= 3X3

CamScanner

8H → 9AH
9H → 05H
AH → 10H

BH → FAH
CH → 0AH

DH → 77H
EH → 03H

FH → AAH

machine code

Assembly to machine code.

SUB CH → 0010 1100
ADD 9H → 0001 1000
ADD 8H → 0001 1000
SUB EH → 0010 1110
SUB EH → 0010 1110
SUB EH → 0010 1110
OUT → 1101 XXXX
HLT → 1011 XXXX

(b)

AX	BX
5 − 10 = −5 H	AH
−5 + 9A = 95 H	9AH
95 + 9A = 12F H	9AH
12F − 3 = 12C H	3 H
12C − 3 = 129 H	3 H
129 − 3 = 126 H	3 H
126 H	3 H
0 H	0 H

Fall 25

(a)

initially, AX = 4H
BX = 1H
Output Reg = 3H.

op code table থেকে operation নেব

machine code to Assembly

0H → 0011 1110 → SUB EH
1H → 0100 1001 → ADD 9H
2H → 0001 1000 → OUT
3H → 0100 1100 → ADD CH
4H → 1100 1101 → LDA DH
5H → 0011 1011 → SUB BH
6H → 1000 1110 → HLT
7H → 0100 1010 → ADD AH

AX	BX
4 − 2 = 2	2
2 + 41 = 43	41
43	41
43 + 4 = 47	4
60	4
60 − 21 = 39	21
0	0
0	0

HLT মানে comp off মানে এ

8H → Decimal 79 AH → 7 CH → 4 EH → 2

9H → 41 BH → 21 DH → 60 FH → 31

(d) $-3-5\times4$

$= [-3-5-5-5-5]$ ✗ → ভুল execute করছে

$= [-3-4-4-4-4-4]$ ✓ → ভাল Data তেই RAM এ। তাই efficient হওয়া সত্ত্বেও বাদ।

	AX	BX
SUB AH	4−7 = −3	7
SUB CH	−3−4 = −7	4
SUB CH	−7−4 = −11	4
SUB CH	−11−4 = −15	4
SUB CH	−15−4 = −19	4
SUB CH	−19−4 = −23	4
OUT	−23	4
HLT	0	0

Summer 25

$9+1-2\div 1/3$

$= 9+1-2\times 3$

$= [9+1-3-3]$ → efficient ✓ (RAM এ BH এ 3 আছে তাই- করা যাবে)

$= 9+1-2-2-2$

	AX	BX
ADD FH	5+4=9	4
ADD 8H	9+1 =10	1
SUB BH	10−3 = 7	3
SUB BH	7−3 = 4	3
[HLT 0 0]		

n
সব zero. OUT then HLT

Spring 25

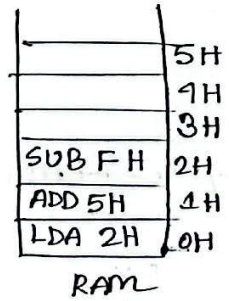
$(3+1) \times 2 - 5$

$= (3+1+3+1) - 5$

		AX	BX
0011 1111 ←	LDA FH	3	
	ADD DH	3+1=4	
	ADD FH	4+3=7	
	ADD DH	7+1=8	
	SUB BH	8-5=3	
1101 XXXX ←	OUT	3	
1011 XXXX ←	HLT	0	

MAR = memory address Reg
↳ connecting medium.

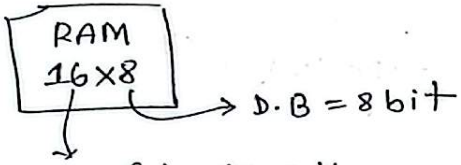
Ins. Reg = এক এক্সটা ins. কে hold করে রাখে।



PC = 0 ←

↓

মানে 0H address এর instruction টাকে execute করতেছে।



No. of blocks = 16

$\Rightarrow 2^n = 16$

$\Rightarrow 2^n = 2^4$

$\Rightarrow$ A.B = 4 bit

PC $\Rightarrow$ 4 bit

MAR $\Rightarrow$ 4 bit

Ins. Reg $\Rightarrow$ 8 bit.

AX = 8 bit

BX = 8 bit

output Reg = 8 bit.



Lecture-06
19/09/26.

$\overline{Su}$ = 0(sub)
= 1(add)

Category : Diagram related.

E → Enable (Device → Bus)
L → Load (Bus → Device)

Register Characters

C_P = 0 (The value of pc no change)
= 1 (PC++)

E_P = 0 (Inactive)
= 1 (PC → Bus)

$\overline{L_M}$ = 1 (Inactive)
= 0 (Bus → MAR)

chip ← $\overline{CE}$ = 1 (Inactive)
= 0 (RAM → Bus)

$\overline{L_I}$ = 1 (Inactive)
= 0 (Bus → instruction Reg)

$\overline{E_I}$ = 1 (Inactive)
= 0 (instruction Reg → Bus)

$\overline{L_A}$ = 1 (Inactive)
= 0 (Bus → Accumulator)

E_A = 0 (Inactive)
= 1 (Accumulator → Bus)

Su (unit)
S_u = 1 (sub)
= 0 (Add)

E_u = 0 (Inactive)
= 1 (Adder/sub Tracker → Bus)

$\overline{L_B}$ = 1 (Inactive)
= 0 (Bus → B register)

$\overline{L_O}$ = 1 (inactive)
= 0 (Bus → output register)

কোন সময় কোন device এর pin কে active বা inactive করবো তা decide করে control sequence

$CON = C_P E_P \overline{L_M} \overline{CE} \cdot \overline{L_I} \overline{E_I} \overline{L_A} E_A \; S_u E_u \overline{L_B} \overline{L_o}$

$= 0011 \quad 1110 \quad 0011$ [device এখন কোন কাজে ব্যবহৃত হচ্ছে না। তাই inactive]

$= 3E3H$

SAP তে 6 টা cycle লাগে [প্রথম 3টা cycle এ fetch করে পরের 3টা cycle এ execute করে]

LDA instruction:

T1: 1. PC → Bus (E_P) | এটার জন্য $CON = C_P E_P \overline{L_M} \overline{CE} \; \overline{L_I} \overline{E_I} \overline{L_A} E_A \; S_u E_u \overline{L_B} \overline{L}$

2. Bus → MAR ($\overline{L_M}$) $= 0101 \quad 1110 \quad 0011$

$= 5E3H$

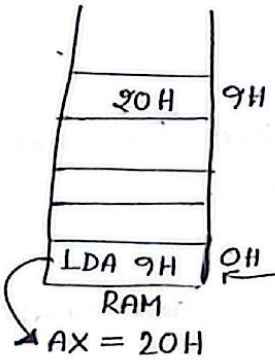
T2: 1. PC++ (C_P) $CON = 1011 \quad 1110 \quad 0011$

$= BE3H$

T3: 1. RAM → Bus ($\overline{CE}$)

2. Bus → instruction Reg. ($\overline{L_I}$) $CON = 0010 \quad 0110 \quad 0011$

$= 263H$



T1 done

PC=0 → MAR = 0

(প্রত্যেক instruction এর জন্য T_1, T_2, T_3 same)

T_4 :
1. Ins. Reg → Bus ($\overline{E_i}$)
2. Bus → MAR ($\overline{L_M}$)

CON = 0001 1010 0011
= 1A3H.

T_5 : 1. RAM → Bus ($\overline{CE}$)
2. Bus → Accumulator ($\overline{L_A}$)

CON = 0010 1100 0011
= 2C3 H.

T_6 : NO operation (LDA operation এর জন্য) ~~CON~~ -
all pins inactive

CON = 3E3 H.

ADD instruction:

T_1, T_2, T_3 same

T_4 : 1. Ins. Reg → Bus ($\overline{E_i}$)
2. Bus → MAR ($\overline{L_M}$).

T_5 : RAM → Bus ($\overline{CE}$)
Bus → B Reg. ($\overline{L_B}$)

CON = 0010 1110 0001
= 2E1 H

T_6 : 1. ADD (Su = 0)
2. Adder / sub → Bus (E_u)
3. Bus → AX ($\overline{L_A}$)

CON = 0011 1100 0111
= 3C7 H

SUB instruction:

$T_1 - T_5$ (add এর মতো)

T_6 : 1. SUB (Su = 1)
2. Addr/Sub → Bus (E_u)
3. Bus → AX ($\overline{L_A}$)

CamScanner

OUT operation: T_1, T_2, T_3 same

T_4: 1. Accumulator ⟶ Bus (E_A) $CON = 0011\ 1111\ 0010$

2. Bus → output Reg ($\overline{L_O}$) $= 3F2\ H$.

T_5: NO operation. All pins inactive

T_6: NO Operation. All pins inactive

CamScanner

Lecture - 07.

16/09/26 · (wednesday).

problem solving session ☺

Spring 261

Initially,

AX = 5H BX = 10H out Reg = 7H MAR = 9H (address)

CON = $\overline{L_I}\,E_I\,L_A\,\overline{E_A}\,\overline{E_U}\,S_U\,L_O\,L_B\,\overline{E_P}\,C_P\,\overline{L_M}\,\overline{CE}$

(প্রত্যেক binary)

T4: (7EAH) = 0111 1110 1010

T4: L_A & (Bus → AX)

L_O (Bus → out. Reg)

$\overline{CE}$ (RAM → Bus)

T4:	AX	BX	out. Reg
	5H	10H	5H

T5 (ADBH) = 0100 1101 1011 ($\overline{E_A}$, L_B)

T5: $\overline{E_A}$ (AX → Bus)

L_B (Bus → B. Reg)

T5:	AX	BX	out. Reg
	5H	5H	5H

T6 (72BH) = 0111 0010 1011 (L_A, $\overline{E_U}$, S_U, L_O)

T6:	AX	BX	out. Reg
	0H	5H	0H

T6:

L_A (Bus → AX)

$\overline{E_U}$ (Adder6/sub → Bus)

$S_U = 0$ (Sub)

L_O = (Bus → output. Re)

ADD:

(i) T_4 : 1. ins. Reg → Bus ($\bar{E_I}$)
2. Bus → MAR ($\bar{L_M}$)

CON = $E_U S_U \bar{L_O} \bar{L_B}$ $\bar{L_I} \bar{E_I} \bar{L_A} E_A$ $E_P C_P \bar{L_M} CE$

= 0 1 1 1 1 0 1 0 0 0 0 0

= 7A0 H

$E_U S_U L_O L_B$ $L_I E_I L_A E_A$ $E_P C_P I_M CE$
0 1 1 1 1 1 0 0 0 0 1 1

(ii) T_5 : 1. RAM → Bus (CE) CON = 7C3 H
2. Bus → AX ($\bar{L_A}$)

T_6 : 1. AX → Bus (E_A) 0101 1111 0010
2. Bus → out. Reg ($\bar{L_O}$) CON = 5F2 H

Summer 252

CON = $\bar{L_I} \bar{E_I} \bar{L_A} E_A$ $E_P C_P \bar{L_M} CE$ $E_U S_U \bar{L_O} \bar{L_B}$

T_4(A07H) = 1 0 1 0 0 0 0 0 0 1 1 1

T_4 : $\bar{E_I}$ (Ins. Reg → Bus)
$\bar{L_M}$ (Bus → MAR)

A = 10 E = 14
B = 11 F = 15
C = 12
D = 13

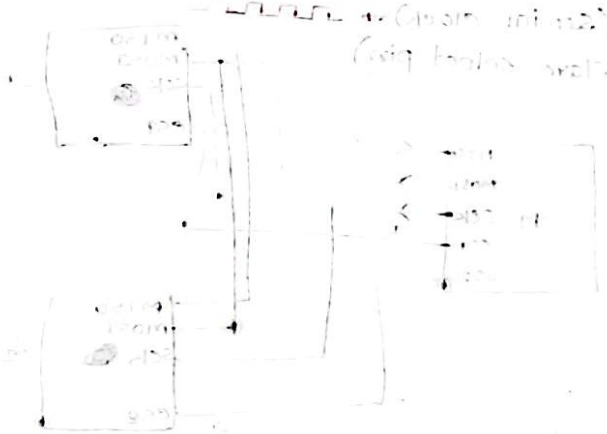
T_5 (E36) = 1110 0011 0110

$\bar{L_B}$ (Bus → B. register)
CE (RAM → Bus)

T_6 (C2F) = 11 $\bar{L_M}$ 00 0010 1111 E_U

$\bar{L_A}$ (Bus → AX)
E_U = (Adder/sub → Bus)
$\bar{S_U}$ = 1 (Add)

Spring 251



Topic: Communications protocols

Serial Communication (cost কম, simple circuit, তবে slow work)
Parallel communication (cost বেশি, complex circuit), fastest)

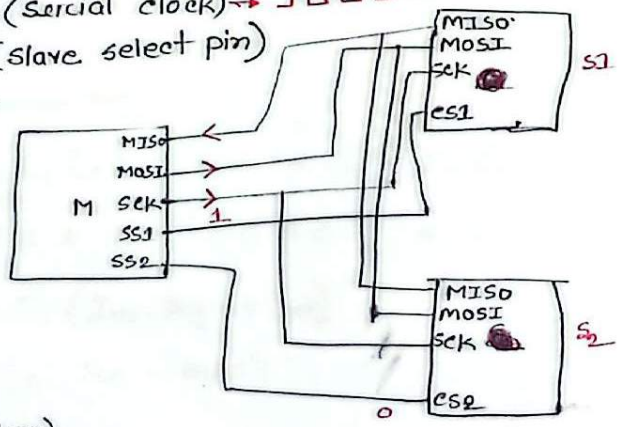
Serial communication
→ SPI
→ I2C [inter - integrated circuit]

SPI: Master device একটাই থাকবে।

total system এ 10টা device থাকলে Master 1 টা slave 9টা।

* Master নষ্ট হয়ে গেলে পুরো communication down.

Pins: MISO (Master In slave out) → Master Data receive
MOSI (Master Out slave IN) → Master Data sent
SCK (Serial clock) → ⎍⎍⎍
SS (Slave select pin)



Master মাস্টার
Slave থাকলে মাস্টার master এ slave select pin অবশ্যই হবে।
System এ 3 MISO, MOSI, SS pin থাকবে। Slave এও থাকবে।

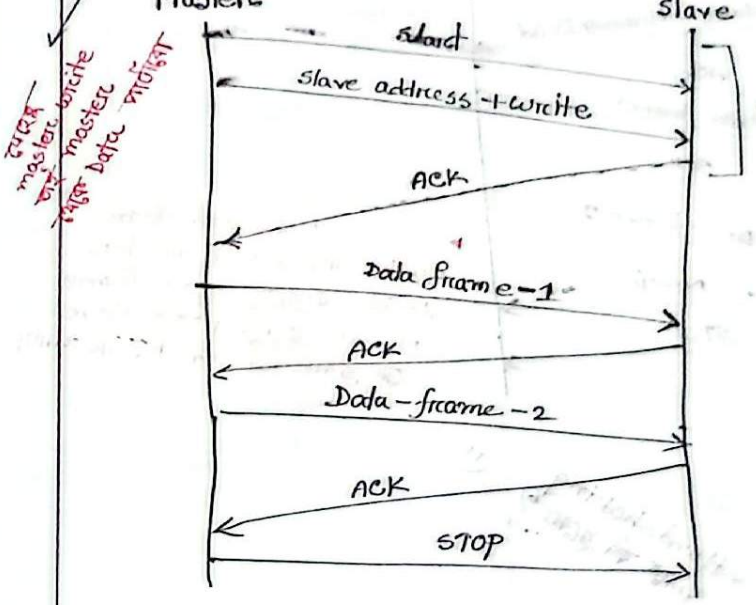
I2C: $\log_2(1000)$
≈ 10.
$= 2^{10}$ → 10 টা pin দিয়ে 1000 টা control করতে পারবো।

Serial communication হয় দুইভাবে করে।

Read = 1
write = 0

* why does I2C is called handshaking protocol?

Master write Diagram:

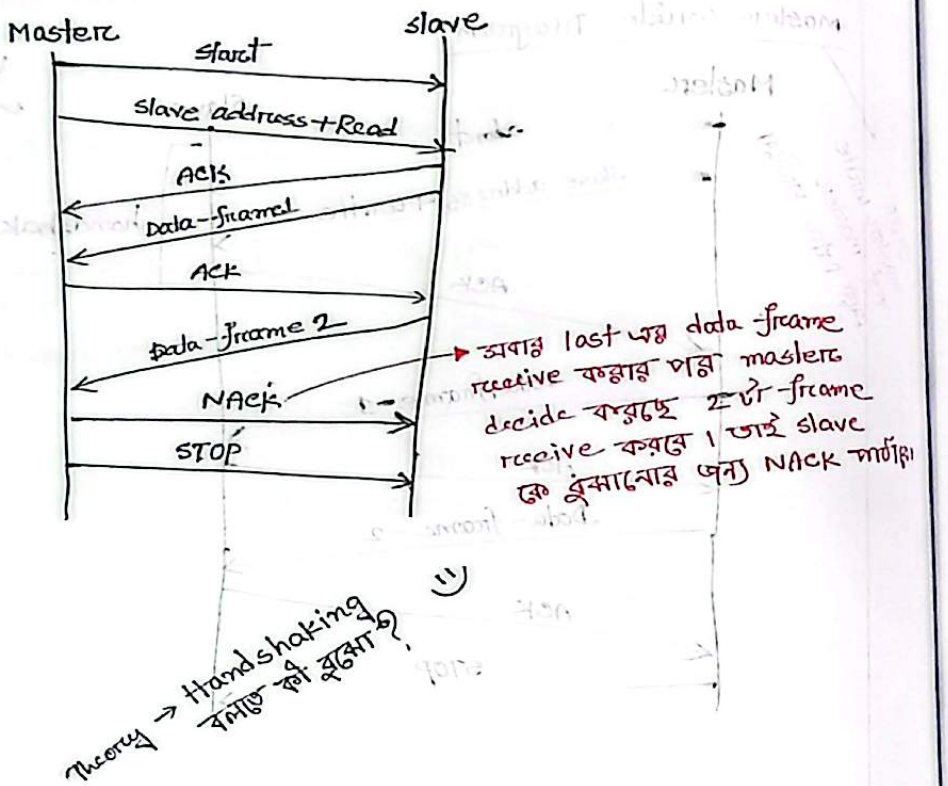


handshaking.

Data frame এ 8 bit করে Data থাকে।

এই operation এ NACK আসবে।

Master Read Diagram :-



receive

এবার last এর data frame receive করার পর master decide করছে 2nd frame receive করবে। তাই slave কে বুঝানোর জন্য NACK পাঠাচ্ছে।

Theory → Handshaking বলতে কী বুঝায়?

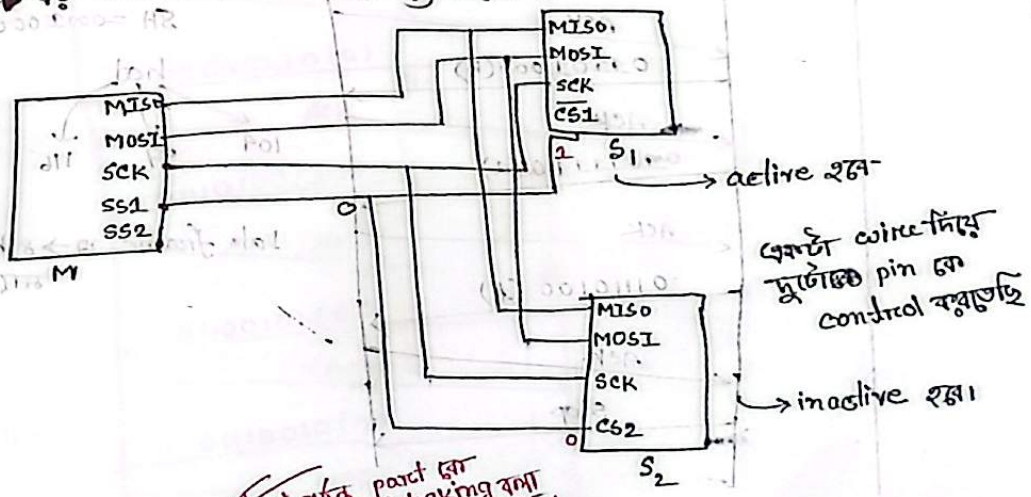
Lecture - 03.
23/09/2026 - (Wednesday)

Topic: Communication protocols:

Serial communication (slower, simple design, less cost
parallel communication (faster, complex design, costly

SPI :→ Master- slave বলা হয়
→ always master device একটাই হবে।
→ Master device নষ্ট হয়ে গেলে পুরো communication down হয়ে যাবে।
↳ small type এর communication এর জন্য badders.

I2C :
↳ বড় communication এর জন্য ভালো।



এই পর্যন্ত part কে handshaking বলা হয়।

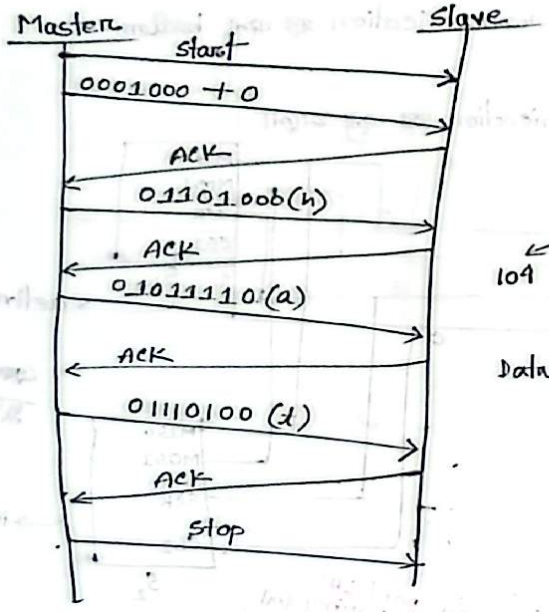
Start → slave এর address পাঠাবে → master বুঝাবে read operation/ write operation করবে
→ slave থেকে ack/Nack আসবে → Data frame এ 8 bit data পাঠাবে
→ ACK/NACK wait করবে → কাজ শেষ হলে stop করবে master.

Slave address হবে 7 or 10 bits

"Questions"

Master (index: 2H) send data (chare 'hat') to the Slave (index: 8). Draw the corresponding sequence Diagram.

Sol^n: → প্রথমে কাজ কোন operation তা বের করবো। এখানে master send data করতেছে তাই master write.



write = 0
Read = 1

SH = 0001000

hat

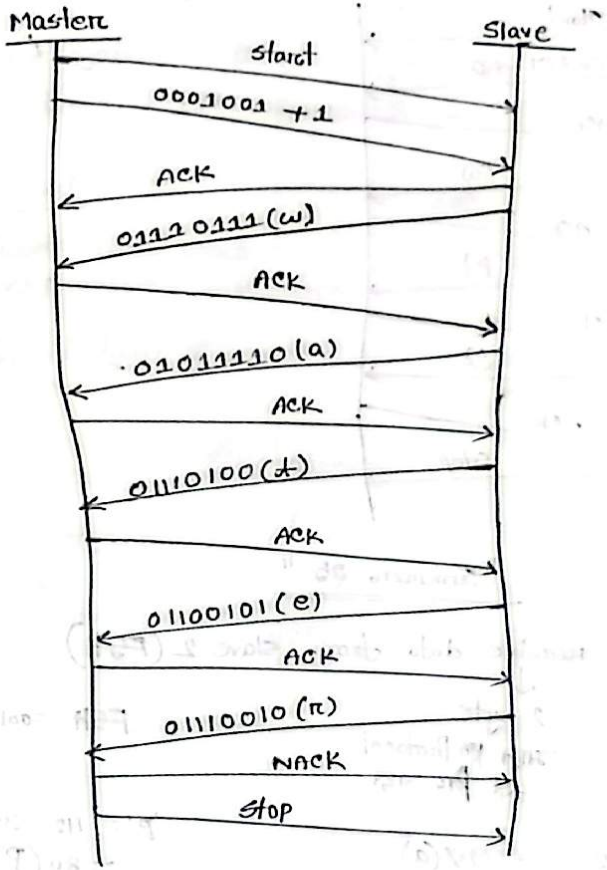
104 97 116

প্রতিটি character 1 Byte করে।

Data frame. এ → 8 bit-এ থাকে মানে 1 Byte.

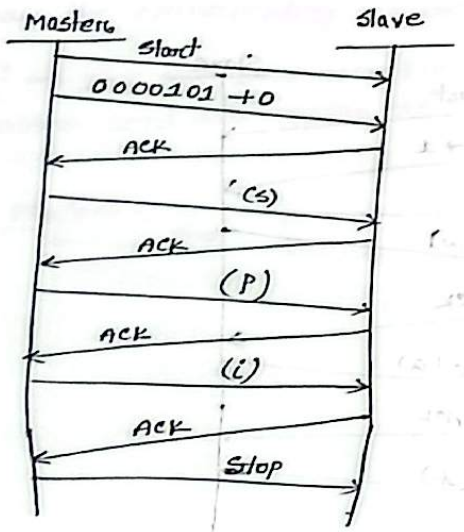
master read operation

Master receive 5 byte data (chare 'watere') from the slave (index: 9H).



"Spring 261"

2) (b) Master write operation.



A → 65, a → 97, b → 98

P

i

"Summer 25"

Master 2 receive data from slave 2 (F5H)

↓

2 byte

মানে ২টা Protocol এর Bit থাকে

FSH = 001111 0101

A → 65 + 32 = 97 (a)

'P' = 112 − 32 = 80 (P)

Scenario based Question থাকবে

Topic: MicroController

Lecture − 09

Thursday (Makeup)

21/09/26

up + storage + I/O → uC

✓ uC নির্দিষ্ট specific কাজের জন্য ব্যবহার করা হয়। একই type এর কাজ বার বার করে।

✓ up base system = Raspberry pi → processor বেশি powerful.

✓ Arduino = uC মানে specific কাজ করে।

up	uC
1. Software based work (Heavy processing, calculation)	1. Hardware based work (Sensors + Actuators)
2. Different types of work	2. Specific or same type of work.

Architecture

instructions থাকে

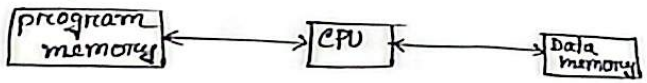


Fig: Harvard Architecture → faster, cost বেশি, complex circuit

Von Neuman Architecture: Memory একটাই। এটাতে Data + instructions একসাথে থাকে। এটা slower, cost কম, simple circuit

processor ও cache বড়ো, RAM ও বড়ো না।

Instructions Set Architecture

→ RISC (Reduced instruction set computers)

→ CISC (complex instruction set computers)

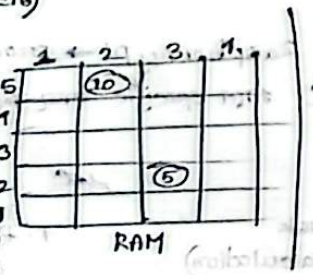
RISC

Load A, 2:3 ⇒ A = 5

Load B, 5:2 ⇒ B = 10

PROD A, B ⇒ A×B = A×B
= 5×10
= 50

STORE 2:3, A



CISC

5×10 = 50

MULT 2:3, 5:2

RISC : Simple command
→ fixed length instruction
→ Register বেশি

CISC : Complex command
→ Variable length instruction
→ Register কম

Math অংশ।

Topic : Analog to digital and Digital to analog conversion

Analog : যেকোনো value নিতে পারে। Continious data. কোনো gap নেই।

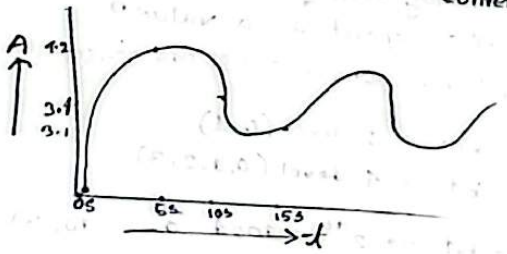
Digital : Discrete Data → যে fixed কয়েকটা value নিতে পারবে।

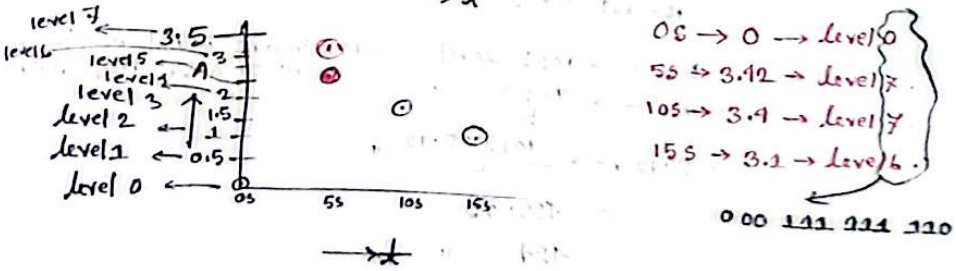
0 1 2 3 → fixed এর বাইরে আর নিতে পারবে না।

ADC [Analog to Digital Conversion]

ADC → PCM ⇒ pulse-code Modulation
→ 1. Sampling (Data কমানো হয়)
→ 2. Quantization (fixed value করা হয়)
→ 3. Encoding (level গুলোকে simply binary তে convert করবে)

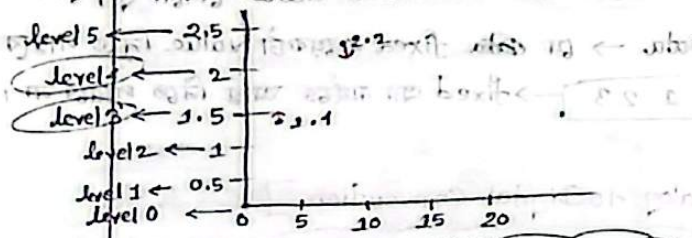
speed → interval বাড়ানো
accuracy → ঐ কমানো





level যত বাড়ানো → Quantization error তত কমে যাবে।

Lecture -10.
27/09/2026.
Sunday !!.



OS → level 0 → 000, 011, 100
5S → level 3
10S → level 1

Question: 10 bit ADC of the Arduino a 5V system. If the analog voltage is 2.12 V then what will be the ADC report as a value?

Soln: ১) level কয়টা হবে জেটা আগে বের করবো।

1 - bit → 2 level (0, 1)
2 - bit → 4 level (0, 1, 2, 3)
!
10 - bit → $2^{10} = 1024$ (0 ..... 1023)

5V → 1023 level → Maximum level maximum value এর সাথে map করবে।
∴ 1V → $\frac{1023}{5}$ "
∴ 2.12V → $\frac{1023 \times 2.12}{5}$ "
$= 433.75$ "
$= 431$ "

or,
using formula, maximum level বসাই

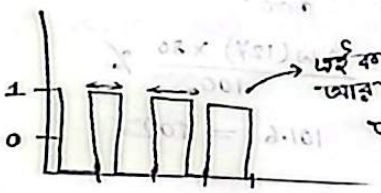
$$\frac{\text{Resolution of the ADC}}{\text{system voltage}} = \frac{\text{ADC reading}}{\text{Analog voltage measured}}$$

$$\Rightarrow \frac{2^{10}-1}{5} = \frac{x}{2.12}$$

$$\Rightarrow x = \frac{1023 \times 2.12}{5} \approx 433.75 \approx 431$$

Digital to analog converter

↳ Technique used : Pulse Width Modulation (PWM)



এই কাজটে আরো fast করলে এটাকে আর disconnected বলে মনে হবে না।

একটা cycle এর মধ্যে কত portion one pass করছে এটাকে বলে Duty cycle.

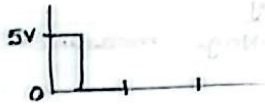
Math:

12 bit digital to analog, analogwrite (1024) ← এটাই বলে দিবে duty cycle কত হবে।

12 - bit → $2^{10} = 4096$ (0 ... 4095)
analogwrite (4095) ⟶ D.C 100%
" (1) ⟶ " $\frac{100}{4095}$ %
" (1024) ⟶ " $\frac{100 \times 1024}{4095}$ %
$= 25$ %

∴ Duty cycle = 25%

10% → 50% এর



"Spring 261"

21 @ 7 bit $= 2^7 = 128$ (0 ··· 127)

$\frac{1}{5} = 0.8 = 80\%$

Duty cycle 10% ∴ A.W (127)

∴ 1 % $= \frac{A.W(127)}{100}$ %

∴ 80 % $= \frac{A.W(127) \times 80}{100}$ %

$= 101.6 = 102$

Fall 253

① @ Tonic Tahmid,

12 - bit = 4096 (0 ··· 4095)

A.W (4095) ··· D.C 100 %

A.W (1) ··· $\frac{100\%}{4095}$

A.W (128) ··· $\frac{100\% \times 128}{4095} = 11.91\%$

Sajib,

11.91% D.C ··· A.W (122)

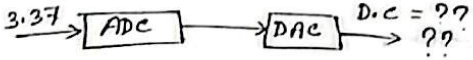
∴ 1% D.C ··· A.W $\left(\frac{122}{11.91}\right)$

∴ 100% D.C ··· A.W $\frac{122 \times 100}{11.91}$

$= 1024$

$= \log_2(1024) = 10$ bit.

ⓑ



8 bit $= 2^8 = 256$ (0 ··· 255)

5 V ··· 255

∴ 1 V ··· $\frac{255}{5}$

∴ 3.37 V ··· $\frac{255 \times 3.37}{5}$

$= 172$

A.W (255) ··· D.C 100 %

A.W (1) ··· D.C ··· $\frac{100\%}{255}$

∴ A.W (172) ··· D.C ··· $\frac{100\% \times 172}{255}$

$= 67.45\%$

"Summer-22"

2. (a)

$$\text{ADC Reading} = \frac{\text{R.of the ADC x Analog vol. M.}}{\text{S.V}}$$

Device A,

$$\text{Adc reading} = \frac{(2^8 - 1) \times 8}{10}$$

Device B,

$$\text{Adc reading} = \frac{(2^{16} - 1) \times 0.4}{x}$$

According to the question,

$$\frac{255 \times 8}{10} = \frac{(2^{16} - 1) \times 0.4}{x}$$

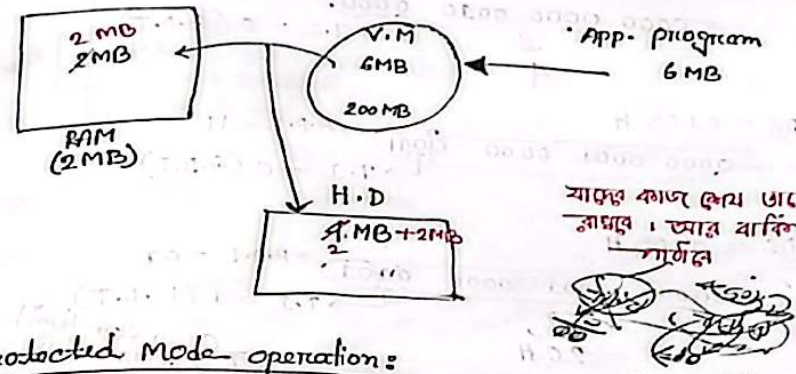
$\Rightarrow x =$ (Ans)

$x = 1.30$ bit

10 marks.

Topic: Memory management and protected Mode addressing

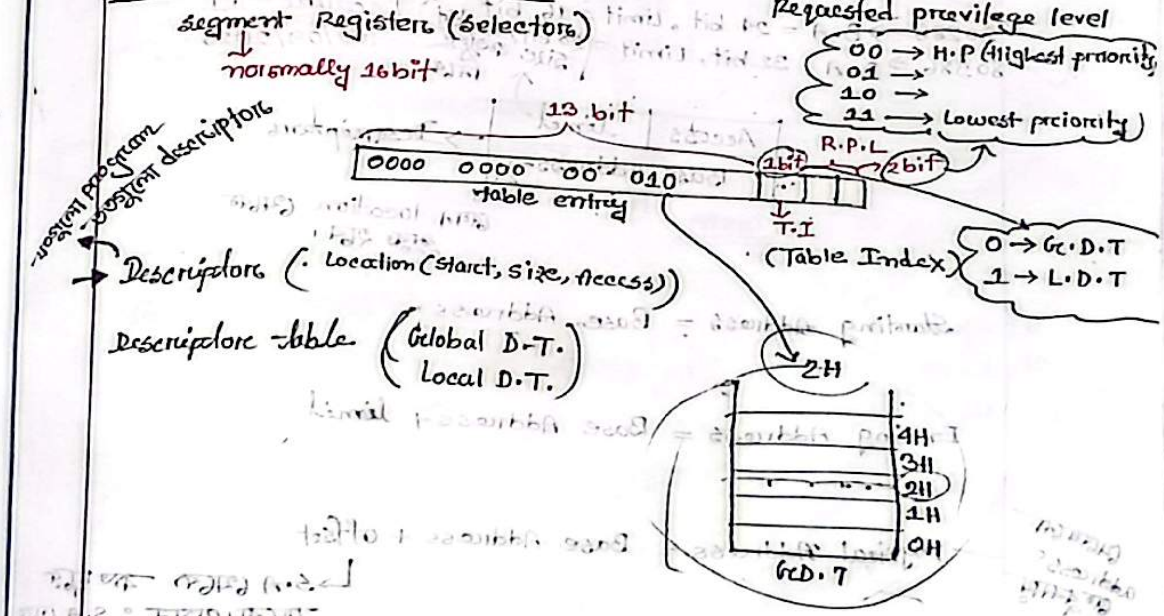
virtual memory (V.M)



protected Mode operation:

Segment Register (Selector)

↓ normally 16 bit



DS = 0020 H

= 0000 0000 0010 0000 → R.P.L
= 4 → T.I = 0 (G.D.T).

DS = 0103 H

= 0000 0001 0000 0011 → R.P.L = 11
→ T.I = 0 (G.D.T)

DS = 0105 H

= 0000 0001 0000 0101 → R.P.L = 01
→ T.I = 1 (L.D.T)
= 20 H

80286 ⇒ B.A = 24 bit, Limit = 16 bit → Gc=0 হলে limit 20 bit, Gc=1 " " 32 bit
80386 ⇒ B.A = 32 bit, Limit = 20 bit/32 bit সাইজ কতটুকু নিবে

Lecture—11
30/09/2026.

Access	limit
Base address	

→ Descriptors

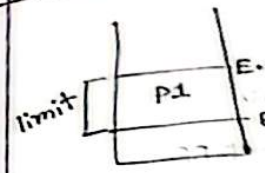
Base address যেখান location থেকে শুরু হবে।

Starting Address = Base Address

Ending Address = Base Address + limit

Physical Address = Base Address + offset

↳ S.A থেকে কত দূরে যাবো। যেমন : S.A থেকে 1 নেব মানে offset 1.



80286 processor, Problem

Given,
offset = 10 H
DS = 3C H

= 0000 0000 0011 1100 → R.P.L
= 3H → L.D.T

S.A = B50000H
E.A = B50000 + OFFF
P.A = B50000 + 10 H (Ans)

Gc = 1 ⇒ limit → 32 bit,
E.A = B.A + Limit * 4k (4k − 1)

Segment register value = 16 bit হয়।

যেখান থেকে শুরু তা Question এর index এ দেওয়া থাকে।

31 bit
2^{31}
$= 2^{21} \times 2^{10}$
$= 16 \times 1K$
$= 16K$

$4K = 4 \times 1K$
$= 1 \times 2^{10}$
$= 2^{12}$

$= 4 \times 2^{10}$
$= 2^{12} = 4K$

30 bit
$= 2^{10} = 1K$

80386 processor:-

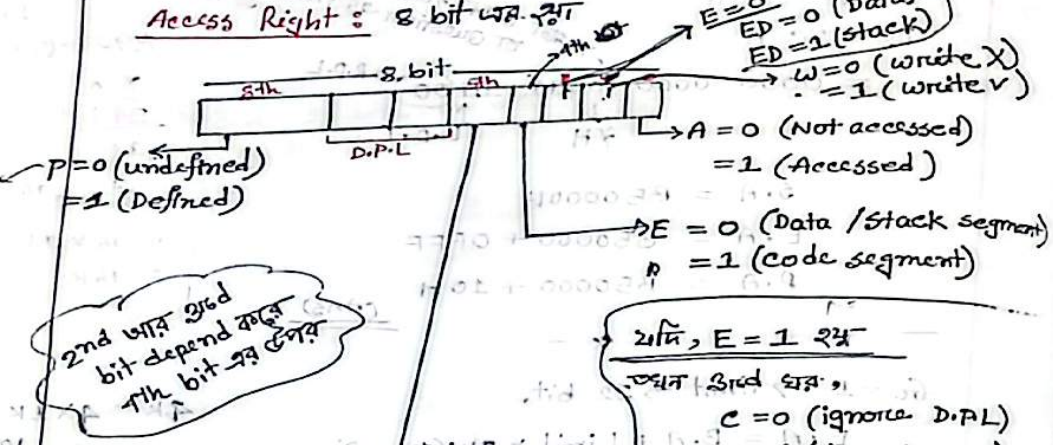
Limit = 001FF ; Gr=1.

∴ Limit * 1K = 001FF * 4K (1000 (Hex)) = 001FF000

Limit * 4K + (4K-1) = 001FF000 + FFF
= 001FF FFF

—— x ——

Access Right : 8 bit থাকে।



যদি, E=1 হয়

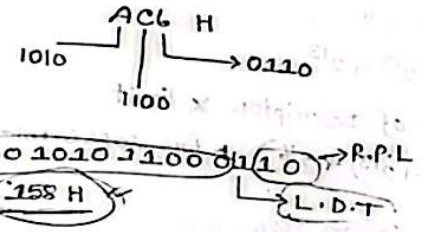
তখন 3rd ধরব,
C=0 (ignore D.PL)
=1 (do not " ")

2nd ধরব,
R=0 (Read X)
R=1 (Read √)

12/10/26 Lecture-12.
Sunday
"1-10-2026"

Question Solve

i)



ii) S.A = B.A
= C002A00E

Gr = 1

E.A = B.A + limit * 4K + (4K-1)
= C002A00E + 2A043 * 1000 H + (1000-1)
= EA06E00D

iii) P.A = B.A + offset
= C002A00E + C051B =

iv) Access right থাকলে কোন segment তাকে বের করা যায়।

∴ Access right = B4 H
1011 → 0100 → A=0, ED=1 (stack), E=0 (Data/Stack)

(v) A=0 (so not accessed)

(vi) P = 1 (Defined)

Fresh

Virtual Memory Size calculation

L.D.T = 2^{13} num of rows

G.D.T = 2^{13} " "

Total descriptors = $2^{13} + 2^{13}$

Total size = (total) num of descriptors × limit

= $(2^{13} + 2^{13}) \times 2^{16}$ → for 80286 micro-processor.

$\frac{80286}{(2^{13}+2^{13}) \times 2^{16}}$

$\frac{80286}{(2^{13}+2^{13}) \times 2^{20}}$

অথবা,

$(2^{13}+2^{13}) \times 2^{32}$

Spring 2026

1) CS = 7C02 H

0111 1100 0000 0010 → R.P.L

F8 0 H → G.D.T

2) G = 1 ; 2KB question ও বলা আছে। → $2 \times 1K = 2 \times 2^{10} = 2^{11}$ → 11 bit-এ change so 11 টা zero আসবে একটা 1

E.A = B.A + limit * 2K + (2K − 1)

= 1F308810 + 23BD7 * 800 H + (800H − 1)

3) 51 H

0101 → 0100 A = 0 (not accessed)

D.P.L = 10

E = 0 (Data/Stack)

ED = 1 (Stack)

4) L.D.T = 2^{13}

G.D.T = 2^{13}

∴ $(2^{13}+2^{13}) \times 2^{31}$

Summer 252

1) 0000 1 0101 1000 010 → Segment register

L.D.T, R.P.L

= AC2 H.

158 H

0 0001 → 1000

0101

B 2

1011 → 0010 A = 0 (not accessed)

D.P.L

E = 0 (D/S)

ED = 0 (Data)

① a) 00001 0101 1000 010 → segment register
= AC2 H (Ans)

158 H
0001 | 1000
0101

b) Starting address = Base address = D6590B00
ending " = B.A + limit = D6590B00 + FB000
= D668BB00 H

c) Physical address = ~~D668BB00~~ D6590B00 + D096E
= D666146E H (Ans)

d) Access = B2 H
→ 0010
→ E = 0
ED = 0 (data)
→ 1011
As A = 0 so segment has not been accessed.

e) E = 0, ED = 0 so Data segment.

f) D.P.L = 01

given, P.A = 92A15BF8 H

index: E82 H → ~~S.A = D721 0111~~
E.A = ~~D721 0111 + 0123 * 4K + (4K - 1)~~

015 H → S.A = 23308810
E.A = 23308810 + 18BD7 * 4K + (4K - 1)
= ~~3BEE01A5~~ H 3BEE080F

∴ 92A15BF8 H does not belong to this range.

1A3 H → S.A = 4FE1DBB8
E.A = 4FE1DBB8 + 2315A H = 4FE71012

∴ 92A15BF8 H does not belong to this range.

2F6 H → S.A = 866125A3
E.A = 866125A3 + 1132F * 4K + (4K - 1)
= 9791 25A2 H

∴ 92A15BF8 H belongs to this range.

so, index number is = 2F6 H.

b) P.A = B.A + offset
⇒ 867003A0 = 866125A3 + offset
⇒ offset = 867003A0 − 866125A3
= EDDFD H.